

# Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

## Computer Vision Engineer

Senior Computer Vision Engineer with 3+ years of experience designing and deploying real-time, safety-critical vision systems on embedded and edge platforms. Proven expertise in classical computer vision and deep learning, including detection, tracking, optical flow, and optimisation of CNNs using FP16/INT8 quantisation and asynchronous inference pipelines. Strong C++17 and Python background with a track record of translating research into deterministic, production-ready systems under strict latency and reliability constraints.

## WORK EXPERIENCE

### Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

#### AI Engineer

- Engineered and deployed real-time CNN-based perception pipelines on edge devices using PyTorch and OpenVINO, achieving 15% F1 improvement and deterministic 24/7 operation under harsh oil-rig conditions.
- Applied FP16 quantisation with structured optimisation meeting strict latency, memory, and stability constraints for safety-critical inference.
- Designed and validated deterministic system behaviour through on-site field testing, performing root-cause analysis of model failures and reducing crash frequency by 67% in live video pipelines.
- Refactored legacy monolith into Kafka-driven, Dockerised microservices, improving fault isolation and increasing MTBF by 35% across edge deployments.
- Designed automated data collection, augmentation, and labeling tools to scale large image datasets, increasing dataset diversity by 40% and cutting manual labeling time by 60%.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.

### LiVNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

#### Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Implemented optical flow-based ego-motion estimation feeding motion representations into CNNs to classify vehicle state in production pipelines.
- Optimised neural networks using quantisation-aware training, FP16/INT8 fixed-point inference, structured pruning, and layer fusion, reducing model size by 50% while meeting strict memory and latency budgets.
- Built asynchronous inference pipelines using OpenVINO to maximise throughput and determinism on Intel VPU edge hardware under real-time constraints.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Developed adversarial-condition training and validation suites simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Built automated labelling, dataset versioning, and performance-tracking pipelines, repurposing detection models for annotation and cutting manual labelling effort by 80%, accelerating iteration cycles.
- Mentored engineers on optimisation trade-offs, deployment best practices, and performance debugging, improving delivery consistency across AI, software, and hardware teams.

## EDUCATION

### Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

#### Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and radiological data for the teleoperation of the Clearpath Husky UGV development platform.
- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes and interfaces integrating perception outputs into a broader robotic system, managing sensor topics, tf2 frames, and message synchronisation for downstream localisation and navigation.
- Integrated perception pipelines into ROS-based robotic workflows, validating end-to-end data flow from sensors to SLAM and navigation components in Gazebo prior to deployment.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS2 prior to real-world deployment.

### Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

## PROJECTS

### ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 3D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam\_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

## SKILLS

- **Perception & SLAM:** 3D Feature Extraction, Convolutional Neural Networks, Data Augmentation, Eigen, LiDAR SLAM, NumPy, OpenCV, Point Cloud Library, ROS2 image\_transport, SAM, Scikit-Image, U-Net, Vision Transformer (ViT), YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, Docker, DVC (Data Version Control), Git, Intel VPUs, Linux, Microsoft Azure, MLFlow
- **Programming & Development:** Bash, C++17, CMake, colcon, Gazebo (Harmonic), Nav2, Pandas, PostgreSQL, Python, rclcpp, rclpy, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy)
- **Deep Learning and Optimisation:** Bayesian Optimisation, DeepSORT, MultiRocket, ONNX, OpenVINO, Optuna, PyTorch, Quantisation-Aware Training, TensorFlow, TensorRT